

AI Ethics Critical Thinking Assignment — Case Study: Amazon's AI Recruiting Tool (2018)

Summary

Amazon developed an automated resume screening tool intended to rate job applicants for technical roles. After training on roughly ten years of company resumes predominantly from male applicants, the model learned to prefer male candidates and penalize résumés containing words related to women (e.g., “women’s”). In 2018 Amazon abandoned the project after engineers found the system was biased against female applicants and was not fixable by simple rewrites. [reuters.com+1](#)

Critical Analysis - Three Ethical Issues

1) Training-data bias and downstream discrimination

Why chosen: The root cause reported was biased training data: historical hiring at Amazon skewed male, so the model learned gendered hiring patterns rather than merit. This makes it a clear, instructive example of how historical inequities can be encoded into ML systems. [reuters.com](#)

Thought process / implications: At first glance the failure looks like a simple classification error; deeper analysis shows that supervised ML will reproduce correlations present in input data. When hiring data reflect systemic bias (fewer women in technical roles), a model optimizing for “what we hired in the past” will perpetuate and even amplify inequities. Such biased outputs have legal, reputational, and social harms: denied opportunities, reduced diversity, and reinforcement of discriminatory labor markets. Moreover, relying on an opaque score can conceal discriminatory behavior from human reviewers who trust the algorithm. [American Civil Liberties Union+1](#)

Proposed mitigation: Use curated, de-biased training sets or reweight samples to reflect desired fairness criteria; combine algorithmic scoring with rule-based filters (e.g., remove gendered features), and run fairness audits (statistical parity, disparate impact) before deployment. Employ synthetic augmentation to increase representation of under-represented groups when lawful and appropriate. External third-party bias audits should be mandated prior to production use. [Thomson Reuters](#)

2) Lack of transparency and accountability

Why chosen: Amazon initially developed the tool internally without transparent disclosure about its use, criteria, or safeguards raising issues of accountability and informed oversight. [reuters.com](https://www.reuters.com)

Thought process / implications: Opaque systems prevent stakeholders (applicants, HR, regulators) from understanding or contesting decisions. Without transparency, it's difficult to diagnose bias or to implement corrections. Accountability gaps also create organizational risk: when errors occur, responsibility is diffuse. Engineers, product managers, or HR may deflect responsibility. This undermines trust and can lead to regulatory backlash. [reuters.com](https://www.reuters.com)

Proposed mitigation: Adopt “model cards” and documentation for datasets, intended use, limitations, and performance across demographic groups. Require sign-off by a cross-functional ethics board before deployment. Maintain logs and decision explainability tools (local explanations, feature importance) so hires can be audited and contested. Public disclosure of high-level policies (not private IP) builds public trust. [reuters.com](https://www.reuters.com)

3) Overreliance on automation and insufficient human-in-the-loop design

Why chosen: The tool was intended to automate résumé review; engineers found the model's rankings seeped into hiring practice, risking automation bias where humans overly defer to algorithmic suggestions. [reuters.com](https://www.reuters.com)

Thought process / implications: Automation can increase throughput but also creates brittle workflows: hiring teams may reduce diverse candidate consideration if an algorithm deprioritizes them. Overreliance also reduces opportunities for human judgment to catch nuanced contexts (career gaps for caregiving, nontraditional STEM pathways). The ethical risk is not only bias but systemic exclusion built into operational practices. [IMD Business School](https://www.imd.edu/business-school)

Proposed mitigation: Design decision-support systems (not decision-making) where algorithms surface suggestions but humans retain final authority, supported by fairness alerts and mandated blind reviews. Train HR personnel in algorithmic literacy to avoid automation complacency. Measure human-algorithm interplay to ensure human reviewers do not adopt algorithmic biases (periodic randomized audits). [reuters.com](https://www.reuters.com)

Reflection on Critical Thinking

Working through this case shifted my view from thinking bias is a solvable “data cleaning” issue to appreciating it as a sociotechnical problem requiring organizational change. Initially I assumed more data or better models would fix unfair outcomes; after analysis, I see that without deliberate design choices (reweighting, auditing, governance), ML will echo structural inequities. This changed assumption, “more data = better fairness”, now shapes my approach: I will prioritize dataset provenance, demographic impact testing, and cross-functional governance before modeling. Going forward, I will insist on (1) documented model cards and fairness metrics as part of project sign-off, (2) human-in-the-loop workflows with anti-automation training, and (3) independent bias audits for high-stakes tools. This ensures technical rigor is paired with ethical accountability.

Sources (used to validate facts)

- Dastin, J. (2018). *Amazon scraps secret AI recruiting tool that showed bias against women*. Reuters. [reuters.com](https://www.reuters.com/article/ai-recruiting/amazon-scraps-secret-ai-recruiting-tool-that-showed-bias-against-women-idUSKBN1ZK08G)
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- Incident Database report: *Amazon had to scrap its AI hiring tool because it was 'sexist'*. incidentdatabase.ai
- The Guardian. *Why tech's gender problem is nothing new*. (Oct 2018). [The Guardian](https://www.theguardian.com/technology/2018/oct/11/why-techs-gender-problem-is-nothing-new)
- Thomson Reuters / commentary on AI misuse and governance. [reuters.com](https://www.reuters.com)

Notes on AI Use

I consulted news and analysis sources via web search to validate the incident and used those sources to inform the analysis. (No generative AI produced the substantive analysis; the reasoning and reflection are my own.)